

GREATER GAINS IN STRENGTH AND POWER WITH INTRA-SET REST INTERVALS IN HYPERTROPHIC TRAINING

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Abstract

We sought to determine if hypertrophic training with intra-set rest intervals (ISR) produced greater gains in power compared to traditional (TRD) hypertrophic training. Twenty-two males (25 ± 5 yrs, 179.71 ± 5.04 cm, 82.1 ± 10.6 kg, 6.5 ± 4.5 yrs training) matched according to baseline characteristics were assigned to 12 weeks training using TRD or ISR. Body composition, strength (1RM bench and squat), and power output (60% 1RM bench and squat, and vertical jump) were assessed at baseline, 4, 8, and 12 weeks. Determination of myosin heavy chain (MHC) percentage from the vastus lateralis was performed pre and post training. Body composition was analyzed by ANOVA, while performance measures and MHC were analyzed by ANCOVA with baseline values as the covariate. Data are presented as means $\pm$ SD changes pre to post. ISR produced greater power output in bench (TRD 32.8 ± 53.4 W; ISR 83.0 ± 49.9 W, $p=0.020$) and vertical jump (TRD 91.6 ± 59.8 W; ISR 147.7 ± 52.0 W; $p=0.036$) with squat power approaching significance (TRD 204.9 ± 70.2 W; ISR 282.1 ± 104.2 W; $p=0.053$) after post hoc analysis ($p < 0.10$). ISR produced greater gains in bench (TRD 9.1 ± 3.7 kg; ISR 15.1 ± 8.3 kg; $p=0.010$) and squat (TRD 48.5 ± 17.4 kg; ISR 63.8 ± 12.0 kg; $p=0.002$) strength. Both protocols produced significant gains in lean mass with no significant differences between groups (1.6 ± 2.1 kg; $p=0.869$). MHC_{IIX} percentage decreased ($-31.0 \pm 24.5\%$; $p = 0.001$) while MHC_{IIA} percentage increased ($28.9 \pm 28.5\%$; $p=0.001$) with no significant differences between groups. Results indicate that hypertrophy training with ISR produces greater gains in strength and power, with similar gains in lean mass and MHC alterations as TRD. ISR may be best used in hypertrophic training for strength and power sports.

Introduction

The velocity of contraction decreases over the course of performance of successive repetitions (21). In sports where the ability to generate power is a necessary aspect of performance, the resultant decrease in velocity is counter to the principle of specificity. One method for counteracting the aforementioned phenomenon is to reduce the number of repetitions performed at a given percentage repetition maximum (%RM). While reducing the number of repetitions enables less reduction in power output over successive sets (35), to achieve the same volume of training this method extends the total training time. This is a consideration to both individuals and strength coaches as time with athletes is often regulated. Other methods for the maintenance of velocity, and thus power output, include inter-repetition rest, intra-set rest (27) and cluster sets (16, 17). Inter-repetition rest training involves a brief rest between each repetition, whereas intra-set rest incorporates rest between groups of repetitions (27). Cluster sets is a more specific form of inter-repetition rest in which only a 10-30 second rest is employed between each repetition or group of repetitions within a set (16, 17).

Previous studies comparing traditional (4 set x 6 repetitions with 260 seconds rest) and intra-set rest (8 sets of 3 with 113 seconds rest) utilizing intensities greater than or equal to 85% 1RM in upper body exercises have demonstrated intra-set rest intervals do not provide a benefit over training with traditional rest (13, 27). However, lower body training with cluster sets has demonstrated a tendency toward greater gains in measures of power output (19), while intra-set rest has demonstrated significantly greater gains in power output compared to traditional rest between sets in athletes (22). The only study demonstrating significantly greater gains on power

output was performed over a periodized training period to include phases of hypertrophy, strength and power.

Izquierdo et al. (21) demonstrated the reduction in average velocity was similar over all intensities examined using intensities prescribed for the development of hypertrophy. While the process of hypertrophy is no doubt multi-factorial, the recommended intensity for the development of hypertrophy (4) corresponds with that for the development of muscular power in trained subjects performing multi-joint exercises (24). Furthermore, it has been demonstrated the total volume load and intensity at which that volume load is performed are the primary variables associated with optimal gains in lean mass (10). Thus, if total volume load is equated, intra-set rest may allow for greater velocity training, resulting in greater improvement in power over traditional rest while producing similar gains in lean mass. Despite this evidence, the effect of intra-set rest intervals in a periodized resistance training program designed specifically to elicit hypertrophy has not been evaluated. Therefore, the purpose of our investigation was to determine if intra-set rest hypertrophy training, equated for total volume load and rest, improved power more than traditional hypertrophy training. We hypothesized intra-set rest intervals would result in greater gains in power output compared to traditional rest, while both training interventions would produce similar gains in strength and lean mass.

While it is well established resistance training results in a shift in myosin heavy chain (MHC) composition, primarily an increase in percentage of MHC_{IIA} and $\text{MHC}_{\text{IIA/X}}$ hybrids with a concomitant decrease in MHC_{IIX} (10), recent evidence suggests changes in MHC composition may be velocity dependent. Liu et al. (29) demonstrated a shift in MHC_{slow} to MHC_{IIA} with no change in MHC_{IIX} when performing traditional resistance training followed by ballistic exercise. Higher peak velocities have been reported using intra-set rest intervals (17). However, whether

or not differential effects manifest in MHC composition as a result of training with intra-set rest intervals has yet to be determined. Therefore, a secondary purpose of our investigation was to evaluate changes in myosin heavy chain (MHC) composition as a result of intra-set rest intervals. We hypothesized intra-set rest intervals would result in less reduction in MHC_{IIX} percentage compared with traditional rest.

Methods

Experimental Design and Approach to Problem

A longitudinal research design was employed to compare the effects of traditional rest and intra-set rest intervals in a program designed to elicit hypertrophy. To eliminate any possible confounding factors, the type and order of exercises performed as well as the volume load and total rest were equated between groups. This was critical to the design of the study as others have shown variations in these variables to impact training adaptations (10). After baseline testing, subjects were matched according to baseline physical and performance characteristics and randomly assigned to perform hypertrophic training using traditional rest (TRD) or intra-set rest (ISR) within a periodized hypertrophic training program. Body composition, strength and power output of the upper and lower body musculature were assessed at baseline, after 4, 8, and 12 weeks of training. Dietary intake was reported and analysis conducted prior to each testing session. Muscle biopsies for MHC content were performed at baseline and at the conclusion of the training program.

Subjects

Twenty-two males completed this study. Selection criteria included 1.) males between the ages of 20 and 35; 2.) having at least 2 years resistance training experience to include upper

and lower body at least once per week; and 3.) reporting not having consumed any nutritional or ergogenic supplements excluding protein supplementation and/or a daily vitamin for the previous 6-week period. Those meeting entry criteria were asked to fill out a medical history questionnaire to eliminate those with any possible contraindications to exercise. Subjects meeting all criteria were informed of the experimental procedures and asked to sign an informed consent approved by the Institutional Review Board. Baseline characteristics for the 22 subjects ($n = 22$) are presented in Table 1.

Of the 22 subjects, 45% ($n = 10$) were active duty military, 41% ($n = 9$) were members of the Naval Reserve Officer Training Corps (NROTC) Unit, while the remaining 14% ($n = 3$) were not involved in military operations. All subjects reported participating in resistance training of both upper and lower body at least once a week for the previous two years. Evaluation of self-report logs demonstrated subjects primarily engaged in resistance training for the development of strength and hypertrophy. Additionally subjects reported having consistently engaged in endurance (primarily running) and high intensity interval training as part of their training prior to participation in training intervention. None of the subjects were competitive weightlifters.

[Insert Table 1 about here]

Testing Sessions

Figure 1 provides a summary of testing and training procedures. Prior to baseline testing, subjects underwent an exercise familiarization session led by a National Strength and Conditioning Association Certified Strength and Conditioning Specialist (CSCS) to demonstrate proficiency in all lifts utilized for training and testing. During the same week, subjects attended a nutrition seminar detailing macronutrient guidelines for athletes involved in a resistance training program (5) and instruction on proper dietary intake recording provided by a registered

dietitian. Prior to each testing session, subjects completed a dietary record to include 3 weekdays and 1 weekend day. Dietary recording was completed one week prior to each testing session corresponding to the highest volume and intensity of training during each four week cycle (Figure 1). All food logs were reviewed by a registered dietitian and analyzed using dietary analysis software (*ESHA Food Processor Version 8.6, Salem, OR*). Post workout supplementation (20 g protein, 45 g carbohydrates, 3.5 g fat) was provided on training days (*Muscle Milk Collegiate, Cytosport, Benicia, CA*) for all subjects. In the week of testing sessions, subjects reported to the laboratory having fasted for at least 10 hours for body composition measurement. Height and body mass were recorded to the nearest .01 cm and .02 kg respectively using a self-calibrating digital scale (*Healthometer, Bridgeview, IL*) in socks or bare feet. Body composition was then determined using dual x-ray absorptiometry (DEXA) (*Hologic Discovery W DXA software version 12.1, Waltham, MA*) calibrated according to manufacturer's guidelines and performed by a trained technician. Previous studies indicate DEXA to be an accurate and reliable means to assess changes in body composition (3). Subjects had their strength and power output assessed according to standardized procedures outlined below. Muscle biopsies were obtained prior to baseline testing and at the conclusion of the exercise intervention.

[Insert Figure 1 about here]

Strength Testing

Upper and lower body strength was assessed using 1RM parallel back squat ($1RM_{BS}$) and bench press ($1RM_{BP}$) exercises. Subjects reported to the laboratory having refrained from any exercise outside of daily living for at least 72 hours prior to baseline testing and at least 48 hours prior to testing in weeks 4, 8 and 12. The initial progression strategy for 1RM determination of

both exercises was estimated from self-reporting. Subsequent 1RM progression strategies were based on training logs kept throughout the study. A dynamic warm up lasting approximately 8-10 minutes was performed prior to 1RM determination. Two warm up sets of 5 repetitions at 40-60% 1RM separated by two minutes rest were followed by a three minute rest period and one to two sets of 2-3 repetitions at a load corresponding to 60-80% 1RM. Subjects then began performing sets of 1 repetition of increasing weight for 1RM determination. Three to five minutes rest was provided between each successive attempt. All 1RM determinations were made within 3-5 attempts. Subjects were required to reach parallel in the $1RM_{BS}$ for an attempt to be considered successful as determined by a CSCS certified individual providing a verbal “up” command. $1RM_{BP}$ was considered successful if subject remained in contact with the bench during the entire concentric phase of the lift. $1RM_{BS}$ testing was conducted prior to $1RM_{BP}$ separated by a five minute rest period. The same 1RM testing procedure was utilized for both exercises. All strength testing took place on an Optima Smith Machine (*LifeFitness, Schiller Park, IL*) without counterbalance technology. Foot placement was recorded during baseline $1RM_{BS}$ testing and hand placement was recorded at baseline $1RM_{BP}$ testing. These measurements were then used in subsequent testing for strength and power. All testing sessions were supervised by two CSCS certified individuals to determine success during each attempt.

Power Testing

Power testing commenced at least 48 hours post 1RM testing. Subjects performed the same warm up prior to initiation of power testing. Body mass and reach height were recorded for calculation of power as determined by vertical jump. Two countermovement vertical jumps (CMJ) using less than maximal effort were allowed prior to testing. Three maximum effort CMJ were then recorded separated by two minutes rest. If the third attempt was greater than the first

two, another attempt was allowed until a decrease in jump height was observed with no more than five maximum CMJ allowed. Reach height and jump height were recorded using a commercially available Vertec system (*Sports Imports, Columbus, OH*). The maximum attempt of record was later converted to power in watts (PWR_{VJ}) using previously described procedures (38).

After vertical jump testing, power output was assessed during the concentric phase of the parallel back squat (PWR_{BS}) and bench press (PWR_{BP}) exercises using a relative load of 60% 1RM per each respective exercise. PWR_{BS} was determined preceding PWR_{BP} separated by five minutes rest. All testing was performed on the same Smith machine used for 1RM testing. Subjects were instructed to perform the concentric phase of each lift as explosively as possible, preceded by the eccentric phase. After warm up sets, 3 sets of 5 repetitions at 40-50% 1RM, subjects began performing single repetitions at 60% 1RM for determination of mean power output. During the parallel back squat, rubber tubing was placed at the parallel point. This position was determined during strength testing and ensured subjects reached the appropriate parallel position for the attempt to be a success. Three attempts were allowed with the best recorded for further analysis. If the third attempt was greater than the first two another attempt was recorded until power output declined with no more than five attempts allowed. Three minutes rest was utilized between successive maximal power attempts. Power output was measured using a linear position transducer (*Tendo Fitrodyne, Tendo Sport Machines, Slovak Republic*) with only bar weight used for determination of average power. The highest average power output was used for statistical analysis. The reliability of the Fitrodyne (*Tendo Fitrodyne, Tendo Sport Machines, Slovak Republic*) has been previously reported (23). During post training

(12 weeks) power output was assessed on both parallel back squat ($APWR_{BS}$) and bench press ($APWR_{BP}$) using loads corresponding to 60% baseline 1RM followed by 60% post training 1RM.

Reliability of Strength and Power Testing

The reproducibility of the methods for determination of strength and power were assessed in two trials separated by 7 days in ten resistance trained males (age 25 ± 5 yrs, 181.48 ± 11.21 cm, 91.3 ± 14.0 kg with 8 ± 5 years 5 ± 1 days per week resistance training experience). After signing consent forms approved by the Institutional Review Board, subjects performed strength and power output assessment according to the previously described procedures. After 7 days, subjects returned to perform the same testing protocol. The reliability statistics are presented in Table 2.

[Insert Table 2 about here]

Training

All training sessions commenced with a dynamic warm up identical to that used for strength and power testing. Four supervised workouts a week were performed in the following sequence; two days on, one day off, two days on, two days off. Training intensity was structured into four week cycles of increasing intensity as described in Figure 1 with an unloading week during each week of testing. All sessions were supervised by trained staff with at least 1 – 2 CSCS certified personnel leading sessions. Subjects were not engage in any other training outside of daily living for the duration of the study.

Throughout the training program, both groups performed the same exercises, in the same order and intensity (Figure 1). Table 3 provides the exercises performed as well as the order in which they were performed. All major lifts tested for strength and power output were performed

on the same apparatus used for determination of 1RM and power output. Subjects were instructed to perform the concentric phase of all major lifts in an explosive manner. Verbal encouragement was provided throughout training. To determine differences between training programs, groups differed on the sets, repetitions and rest in all major lifts performed (Figure 1). Subjects in TRD performed a standard hypertrophic training protocol of 4 sets of 10 repetitions for all major lifts with 120 seconds rest between sets. In accordance with previous studies evaluating ISR (12, 27), each set was divided in half and rest was equated so that the ISR group performed 8 sets of 5 repetitions with 60 seconds rest. This manipulation of rest to work where the set is divided in half using an intra-set rest and equating rest between groups has previously been used for determination of differences between TRD and ISR in strength training (12, 27). Both TRD and ISR performed 3 sets of 10 repetitions with 90 seconds rest for all assistive lifts. The load was reduced if subjects were unable to complete the prescribed number of repetitions. Intensity was increased if subjects were able to complete at least 85% of the total volume on major lifts. Timing of rest was performed using stop watches on all lifts by trained personnel. The total volume load (weight x reps x sets) was grouped according to lifts utilizing the upper and lower body for main lifts and push and pull exercises for assistive exercises for later analysis. The training intensity (total volume load/repetitions performed) was grouped according to upper and lower body for main lifts only. Workout logs were maintained and verified throughout the 12 week period. Compliance throughout the entire 12 week study was 95%.

[Insert Table 3 about here]

Biopsy and Myosin Heavy Chain Analysis

Prior to baseline testing and within 72 hours of post testing, a 100-200 mg muscle biopsy of the vastus lateralis was obtained from the participant's right leg with suction using a 5mm

biopsy needle (*Pelomi Industries, Denmark*) according to a modification of Bergstrom's technique (8) as described by Evans et al. (14). All muscle samples were cleansed of visible fat, connective tissue and blood and immediately frozen in liquid nitrogen (-190°C), then transferred and stored at -80°C until analyzed. Muscle samples were later prepared (6) and MHC analyzed according to previously described procedures (30). Briefly, after samples were loaded (15 µL), electrophoresis was performed at a constant voltage of 200 V for 22 h in a gel system (*R.Shadel, San Francisco, CA*). For stacking gel penetration, the first hour, the voltage was kept at 160V. During electrophoresis, the gel system was kept in a temperature controlled ventilated hood. The gels were subsequently stained with a silver staining kit (*Bio-Rad, Hercules, CA*) according to manufacturer's instructions. The protein bands of each MHC isoform on the silver-stained acrylamide gel were densitometrically digitalized using a digital camera (*Fuji LAS 4000, Fujifilm Life Sciences, Wayne, NJ*). The densitometric values were derived as an integral of the band density and band area (*MultiGauge 3.0, Fujifilm Life Sciences, Wayne, NJ*). The amount of each isoform was expressed as a percentage calculated as $(\text{IntegProtein}/\text{IntegAll}) * 100\%$ where IntegProtein is the densitometric integral of the corresponding protein band, and the IntegAll is the densitometric integral of all isoforms in the sample.

Statistical Analysis

All statistical analysis was performed using SPSS Version 19.0 (Chicago, IL). One way ANOVA was used to determine baseline differences in physical and performance characteristics. A 2 x 4 (group x time) ANOVA was utilized to determine differences in body mass, lean mass, and percent body fat. Macronutrient content was analyzed in a 2 x 4 (group x time) ANOVA. Overall total volume load and training intensity were analyzed by independent t-test. A 2 x 4 (group x time) ANCOVA covaried for baseline values as the covariate was utilized to determine

changes over the training period in $1RM_{BP}$, $1RM_{BS}$, PWR_{BP} , PWR_{BS} , and PWR_{VJ} . Independent t-tests were used to determine the difference between pre- and post- $APWR_{BP}$ and $APWR_{BS}$. A 2×2 (group $\times$ time) repeated measure ANCOVA using baseline values as the covariate was used to determine changes in percentage MHC_{slow} , MHC_{IIA} and MHC_{IIX} . Statistical power for all tests was > 0.82 . Where necessary, post hoc analysis was performed using the Bonferroni correction. Statistical significance was defined as $p \leq 0.05$.

The use of null-hypothesis testing in sports science to assess practical significance may be inadequate (7). As such, the effect size was used in accordance with the scale proposed by Rhea (32) for highly trained individuals to determine the treatment effect of both training programs. Additionally, Batterham and Hopkins' (7) method for determination of magnitude based inference was utilized at 5% for both TRD and ISR to provide a qualitative inference. The difference in absolute change from pre to post between ISR and TRD was also calculated and a qualitative inference was determined on the effect of ISR (7).

Results

Baseline Characteristics

No significant differences were observed at baseline in age, height, body mass or lean mass. Additionally, no significant differences were observed in training status between subjects (Table 1).

Macronutrient Intake

Absolute macronutrient intake, as well as percent total calories and intake relative to body mass are presented in Table 4. No significant between group differences were observed in any of the macronutrients measured. Caloric intake did not change over the course of the study.

Protein intake increased significantly from baseline to 3 weeks with no further increase observed. Carbohydrate intake decreased over the course of the experimental period. No significant changes were observed in absolute fat intake, as a percentage of total calories or relative to body body mass.

[Insert Table 4 about here]

Training Volume Load and Intensity

Total volume load of main lifts (upper and lower body) and assistive exercises (push and pull) is provided in Table 5. There were no significant differences between groups for total volume load of main lifts or assistive exercises. Additionally, there were no significant differences in upper ($p = 0.113$) or lower body ($p = 0.333$) training intensity between TRD and ISR.

[Insert Table 5 about here]

Power Output

Data from power output assessments are presented in Table 6. There were no significant differences between groups in any power measurements at baseline. Subjects in ISR experienced greater gains in PWR_{BP} . A graphical representation of percentage increase in PWR_{BP} is presented in Figure 2A. Similar increases were observed in PWR_{BS} in both groups. A significant interaction was observed with post hoc analysis revealing no significant between group differences at any training time-point. However, the difference between groups at both 8 (ISR, 151.0 ± 74.0 W; TRD, 97.5 ± 60.9 W; $p = 0.084$) and 12 (ISR, 282.1 ± 104.1 W; TRD, 204.9 ± 70.2 W; $p = 0.063$) weeks approached significance with subjects in ISR showing a greater increase. The magnitude of the effect was also greater in ISR as well the qualitative inference

on the effect of ISR demonstrated a likely positive effect (Table 10). Evaluating PWR_{BS} relative to lean mass, subjects in ISR showed a greater increase at both 8 ($p = 0.016$) and 12 ($p = 0.038$) weeks. Although the pattern of increase appeared greater when evaluating percentage change from baseline in PWR_{BS} (Figure 2B), only a time effect was noted with no group or interaction effects observed. Subjects in ISR experienced greater increase in PWR_{VJ} . This corresponded to a significant difference at 12 weeks compared to TRD in absolute (Table 6) and percentage change from baseline (Figure 2C).

[Insert Table 6 and Figure 2 about here]

Percentage change from baseline in $APWR_{BP}$ and $APWR_{BS}$ is presented in Figure 3. Subjects in ISR demonstrated a significantly greater improvement in percentage increase from baseline in $APWR_{BP}$ ($p = 0.012$), corresponding to an absolute increase of 71.0 W compared to 13.5 W for TRD ($p = 0.010$). Both ISR and TRD increased in $APWR_{BS}$ from baseline, no significant differences were observed ($p = 0.496$). However, the magnitude of effect size was greater for the ISR group (Table 10).

[Insert Figure 3 about here]

Muscular Strength

The results of strength testing are presented in Table 7. There were no significant differences between groups observed at baseline in either $1RM_{BP}$ or $1RM_{BS}$. Only subjects in ISR experienced an increase at 4, 8 and 12 weeks. This corresponded to greater increases at 4 (ISR, 6.6 ± 6.6 kg; TRD, -1.4 ± 6.2 kg; $p = 0.012$), 8 (ISR, 9.9 ± 6.8 kg; TRD, 2.9 ± 5.8 kg, $p = 0.016$) and 12 (ISR, 15.1 ± 8.3 kg; TRD, 9.1 ± 3.7 kg; $p = 0.051$) weeks. Percent change in $1RM_{BP}$ from baseline is presented in Figure 4A. Again, only ISR increased at 4, 8 and 12 weeks. Greater

percent increase from baseline was observed in ISR at 4 and 8 weeks ($p = 0.017$ and 0.034 , respectively), with 12 weeks approaching significance ($p = 0.082$).

Subjects in ISR experienced greater increases in $1RM_{BS}$ at both 8 (ISR, 49.7 ± 13.8 kg; TRD, 36.9 ± 13.5 kg; $p = 0.024$) and 12 (ISR, 63.8 ± 12.0 kg; TRD, 48.5 ± 17.4 kg; $p = 0.011$) weeks. Percent change in $1RM_{BS}$ from baseline is presented in Figure 4B. Percent change from baseline demonstrated a significant interaction ($p = 0.048$) in which post hoc analysis revealed an almost significant greater gain in week 8 ($p = 0.066$) and a significant greater gain in week 12 ($p = 0.041$) for subjects in ISR.

[Insert Table 7 and Figure 4 about here]

Body Composition

Results of body composition testing are presented in Table 8. While subjects in TRD experienced an increase in body mass at 4 weeks, no further increase was observed and post hoc analysis revealed no significant difference between groups at any time point. Lean mass increased in both groups from baseline to 4 weeks and showed a continued increase at 8 weeks. No further increases were noted. No between group differences were identified. When evaluating percentage change from baseline, a significant time effect was noted ($p = 0.001$) with no significant group ($p = .124$) or interaction effects ($p = 0.219$) observed. There were no significant changes in percent body fat as measured by DEXA during the twelve week training.

[Insert Table 8 about here]

Myosin Heavy Chain Composition

Studentized t-tests of baseline MHC demonstrated a significantly higher percentage of MHC_{IIX} in the ISR group ($p = 0.023$) compared to TRD. No significant differences were observed at baseline in percentage MHC_{IIA} or MHC_{slow}. Results from myosin heavy chain analysis are presented in Table 9. Both training protocols experienced a significant reduction in MHC_{IIX} after 12 weeks of training, with a concomitant increase in MHC_{IIA}. No interaction or group effects were observed in either MHC_{IIX} or MHC_{IIA}. Evaluating percentage change from baseline, a significant time effect ($p = 0.001$) was noted with both ISR and TRD experiencing a decrease in MHC_{IIX}, $-37.9 \pm 24.1\%$ and $-23.4 \pm 23.8\%$ respectively, with no differences noted between groups. A time effect was also noted in percentage change from baseline in MHC_{IIA} for both ISR and TRD, with both showing an increase, $32.0 \pm 28.8\%$ and $25.4 \pm 29.1\%$ respectively ($p = 0.001$). Again, no interaction or group effect was noted. A small but significant decrease in percentage MHC_{slow} was observed in both groups. However, when evaluating percent change from baseline, this did not reach significance ($p = 0.164$).

[Insert Table 9 about here]

Effect Size and Magnitude

Effect size, magnitude of effect size and qualitative inferences are reported in Table 10. ISR magnitude of effect sizes were all moderate or large in performance measures, while TRD was found to have a small magnitude of effect as defined by Rhea (32) for highly trained individuals in $1RM_{BP}$, PWR_{BP} , PWR_{VJ} . TRD had a greater effect size magnitude for lean mass gain, however, the effect size for both TRD and ISR were small and trivial, respectively. ISR demonstrated greater effect size magnitude in both MHC_{IIX} and MHC_{IIA} (large) compared with TRD, while TRD demonstrated a greater effect size for MHC_{slow}. Qualitative inferences were

greater for IRS at the smallest worthwhile change 5%, which is much larger than expected in highly trained athletes, ISR demonstrated greater qualitative inferences in $1RM_{BS}$, PWR_{BP} , $APWR_{BP}$, $APWR_{BS}$, PWR_{VJ} , MHC_{IIX} and MHC_{IIA} , with TRD demonstrating a greater qualitative inference in lean mass only. Additionally, a likely or very likely positive effect of ISR was demonstrated in all measured variables except lean mass, MHC_{IIA} , and MHC_{slow} .

[Insert Table 10 about here]

DISCUSSION

In the current study, we sought to determine if ISR hypertrophic training resulted in greater power gains than TRD hypertrophic training. We further sought to determine changes in MHC composition as a result of ISR. The major findings of this study were that after 12 weeks 1) ISR resulted in greater power output in the bench press exercise and vertical jump, 2) power output as measured during parallel back squat approached significance compared to TRD, and 3) when normalized to body weight and lean mass, ISR were superior, 4) absolute power difference in the bench press was greater in the bench press exercise with the use of ISR, 5) ISR resulted in greater increases in strength in the bench press and parallel back squat exercises, 6) ISR resulted in similar gains in lean mass, and finally 7) changes in percentage MHC_{IIX} and MHC_{IIA} are similar to those experienced during TRD hypertrophic training.

To effectively compare it was necessary to control for total volume load and training intensity. This was accomplished as no significant between group differences were observed in major lifts or assistance exercises in total volume load or training intensity in major lifts. Additionally, while dietary intake was not controlled no significant between group differences were observed in any macronutrients. Data suggest the loss of calories from carbohydrate intake was made up by the high intake of protein, which met current guidelines (5), as evidenced by

increased protein intake with a concomitant decrease in carbohydrate intake with no change in total calories or fat over the 12 week period.

Our original hypothesis regarding power output was supported only in part due to the fact power difference in the parallel back squat exercise only approached significance. However, the effect size was greater and the magnitude based qualitative inference demonstrated a likely positive effect of ISR. When power output was evaluated using the initial load from baseline, intra-set rest intervals resulted in greater increases in bench press but not in parallel back squat, though the effect size and qualitative inference at 5% were greater for intra-set rest intervals as well as a likely positive magnitude based inference from ISR. In contrast to trends observed in bench press and parallel back squat exercises, the difference in power output of the vertical jump took the full 12 weeks to manifest. Once demonstrated, the difference in change from baseline, magnitude of effect size, and qualitative inference at 5% were greater in ISR.

The role of inorganic phosphate (P_i) in velocity of shortening has recently received considerable attention. Increases in P_i occur during muscle contractions, mainly from the breakdown of phosphocreatine (PCr). Models of cross bridge cycling propose P_i is released in the transition from low force weakly attached states to high force strongly attached states (2). Westerblad et al. (39) suggested this implies that the transition to the high force states is hindered by increased levels of P_i . Increases in adenosine diphosphate (ADP) have also been suggested to occur during repeated contractions, coincident with PCr depletion (33). Westerblad et al. (40) demonstrated that in the fatigued state, the velocity of shortening was slower following a longer tetanus compared to a shorter contraction. The authors suggested this was due to transient increases in ADP. Partial recovery occurred in a matter of seconds thought to be due to rapid removal of ADP by enzyme action or diffusion.

While the exact cause of reduced velocity of shortening is still debated (39), the current study suggests ISR allows for a greater power output after training for hypertrophy. It is therefore possible that ISR in the current study allowed for a partial reduction of either P_i or ADP by enzyme action or diffusion. Furthermore, ISR may have allowed for an almost complete resynthesis of PCr, as after a fatiguing maximum voluntary contraction lasting approximately 54 seconds PCr has been shown to be resynthesized to 67% original concentration in just 2 minutes and 87% after 4 minutes (34). The truncated number of repetitions performed per set, 5 in the current study, would utilize less PCr and result in less metabolite accumulation than more traditional hypertrophic training. While not recorded, it was observed the time to complete the 5 repetitions performed by ISR to be approximately 10-15 seconds, while it took almost twice the time for TRD to complete the desired number of repetitions (20-25 seconds).

Support for the superiority of ISR on power output of the upper and lower body musculature has been provided by acute studies in which greater power output (21.6-25.1%) in the bench press exercise compared to TRD has been demonstrated (28), as well as higher peak velocity in the power clean utilizing cluster sets (17, 20). While these results support the use of ISR and cluster sets in chronic training, long term studies have demonstrated a tendency toward greater gains in power output of the lower body musculature (19), as well as significantly greater gains in power output of the lower body musculature compared to TRD in athletes (22). The results of ISR on upper body power output is controversial, with some reporting no difference (22, 27) and one reporting less (13) improvement in power output of the upper body musculature. In the current investigation, we report greater gains in power output of both the upper and lower body musculature after 12 weeks of training.

Differences between the current study and those reporting no difference or less improvement in power output of the bench press exercise may at least partially explain the differing results. It has been demonstrated peak power occurs between 40-60% 1RM in the bench press utilizing a Smith machine (36) which was the apparatus used in the current study. While loads in the current study were greater (65-75%) than those prescribed for bench press power; they fall closer to the desired range than intensities used in previous investigations in which no difference (22, 27) or less improvement (13) was observed (85% 1RM or greater).

The only study to date utilizing loads similar to the current study using multi-joint exercises was conducted by Izquierdo et al.(22); however, the authors failed to demonstrate differences in power output in the initial phase of training with loads corresponding to 10 RM (~75% 1RM) for the bench press. Greater improvements in power output were not completely realized in the current study until 8 weeks of training. The initial phase in which intensity of the two studies was similar lasted only 6 weeks. Therefore, the differences in power output may not have had an opportunity to be realized. Additionally, assessment of power output by Izquierdo et al. (22) involved only the concentric action beginning from a stop position, compared to our determination which allowed a descent phase corresponding to an eccentric component.

In agreement with previous studies, we demonstrated ISR to be superior in the development of lower body power output. While back squat power only approached significance, the magnitude of effect size was greater and the qualitative inference on the effect of ISR was likely positive. Hansen et al. (19) recently compared TRD vs. cluster sets on strength and power of the lower body musculature in elite rugby union players. Cluster sets were only used in strength and power training involving squat and clean movements. Similar to the current study, calculations of magnitude-based inference demonstrated a likely positive effect of cluster sets in

peak power and peak velocity at 40 kg, and peak velocity at bodyweight during the jump squat. The authors concluded TRD resulted in greater strength improvements while some evidence suggested a possible benefit for ISR in lower body power development.

It has been demonstrated peak power occurs between 50-70% in the parallel back squat exercises utilizing a Smith machine (36). The loads in the current study fall within this range (65-75%). Izquierdo et al. (22) have been the only group to demonstrate significantly greater gains in lower body power output after long term training. Again the study by Izquierdo et al. (22) allowed multiple comparisons in a 16 week training period corresponding to three micro-cycles. During the first 6 weeks, subjects performed 3 sets of 10 (10RM) repetitions or 6 sets of 5 (10RM) repetitions corresponding to a hypertrophic phase. Strength training commenced in week 7 and lasted 5 weeks with subjects performing either 3 sets of 6 (6RM) repetitions or 6 sets of 3 (6RM) repetitions. Training concluded with both groups performing 3 sets of 2-4 repetitions with intensities corresponding to 85-90% 1RM. Differences in power output were not observed until the last testing session. This is unique as there were no differences in training intervention during the last 5 weeks. Based on the current results, adaptations may have occurred during the initial few weeks of training and not been realized until the end of training.

Contrary to our original hypothesis, ISR resulted in greater increases in both bench press and parallel back squat strength. This is the first study to report greater strength gains with ISR. Studies comparing ISR to TRD using multi-joint exercises have demonstrated ISR resulted in smaller (13, 19, 27) or no difference (22) in strength gains compared to TRD. Studies demonstrating smaller strength gains with ISR utilized loads corresponding to $\geq 85\%$ 1RM. At those intensities this may very well be the case as increased recruitment of muscle fibers results in greater stimulation, particularly in the fast MHC_{IIX} fibers, as measured by electromyography

(EMG) techniques. EMG activity has been shown to increase leading up to the sixth repetition in a 6RM (85% 1RM) bench press exercise (25). On the other hand, Burd et al. (9) demonstrated maximal EMG activity occurred half way through (i.e. repetitions 5-6) the performance of a set of leg extensions using a load corresponding to 70% 1RM. The results by Burd et al. (9) suggest when performing exercise with loads in this intensity range, maximal recruitment occurs at approximately 50% completion during the initial set, and performance of repetitions past this point results in reduced activation of fibers which are considered quick to fatigue, primarily MHC_{IIX} . Similar to previous studies comparing TRD to ISR (12, 27), the current study divided the number of repetitions performed in TRD hypertrophic training in half and equated rest utilizing intensities ranging from 65-75% 1RM. This may have resulted in greater neuromuscular activation over consecutive sets compared to TRD which may have contributed to the greater strength gains demonstrated in the current study. However, further research is needed to answer this question conclusively.

Izquierdo et al. (22) demonstrated no significant difference in strength between TRD and ISR over a 16-week periodized training period. Intensities during the first six week cycle corresponded to those used in the current study (10RM or ~75% 1RM). In contrast to the results obtained by Izquierdo et al. (22), we observed greater strength gains with ISR after only four weeks in the bench press exercise. Difference in back squat approached significance after four weeks and reached significance by week 8. While both studies utilized the same relative intensities, subjects in the current study were only participating in the training outlined, whereas subjects in the study by Izquierdo et al. (22) were also participating in additional training to include sport specific and endurance training. This may have contributed to the differing results.

Furthermore, while it is doubtful, the differences in ISR may also have contributed to the divergent findings (1 vs. 2 minutes).

In support of our original hypothesis and in support of Ahtiainen et al. (1), training with TRD and ISR resulted in similar increases in lean mass over the duration of the 12 week training program. While TRD appeared to have greater increases, this did not reach significance when evaluating absolute or percentage change from baseline. In the current study, rest intervals were short and congruent with current recommendations for the development of hypertrophy (4). The use of shorter rest intervals using moderate intensities has been associated with greater acute elevations in growth hormone when compared to longer rest periods using higher intensity loads (26). However, Ahtiainen et al. (1) demonstrated that hormonal and hypertrophic response did not vary when short (2 minutes) or long (5 minutes) rest intervals were utilized in a chronic training program when volume was equated. There were no differences in total volume load over the 12 week training period ($p > 0.05$). These data support the previous work by Ahtiainen et al. (1) demonstrating similar hypertrophic response regardless of rest when total volume load is equated.

Contrary to our original hypothesis, no significant differences were observed between groups in changes in myosin heavy chain percentage after 12 weeks of resistance training. Both groups experienced a significant increase in MHC_{IIA} percentage with a concomitant reduction in MHC_{IIX} percentage. Furthermore, a small decrease in MHC_{slow} was also observed, although when evaluating percentage change from baseline this did not reach significance. It is well established; the increase in lean mass associated with hypertrophic training is accompanied by a shift in MHC isoforms, identified by a decrease in the percentage of MHC_{IIX} fibers with a concomitant increase in $MH_{IIA/X}$ and MHC_{IIA} fibers (10, 15, 31). However, Liu et al. (29)

previously reported strength training combined with ballistic exercise lead to a differential effect on MHC shifts after 12 weeks of training, shifting from percentage MHC_{slow} to MHC_{IIA} . While the training protocol in the current study did not explicitly include ballistic movements, the use of ISR has been shown to result in greater velocity of contraction compared to TRD training (17, 18).

The differences in intensity and length of time of the current protocol and that of Liu et al. (29) may at least partially explain our divergent findings. Liu et al. (29) utilized loads corresponding to a much higher intensity than the one used in the current protocol, 93% 1RM vs. 65-75% 1RM. Additionally, the length of training of the current study was twice that of the study by Liu et al. (11). Both protocols fall within the time course of adaptations as have been previously reported (37). Claflin et al. (11) recently suggested an alternative explanation must be responsible for the enhanced fiber function as a result of high-velocity training as they did not find any differences at the single fiber level in size, force or power of type II fibers after 14 weeks of training with high-velocities. While Claflin et al. (11) did not directly assess changes in MHC_{IIA} and MHC_{IIX} percentage, the results from the current study and that of Claflin et al. (11) suggest that the increased performance evidenced as a result of high-velocity contractions may in fact result from neural adaptations rather than physiological adaptations.

Practical Applications

The results from the current study support the use of ISR during training for muscle hypertrophy. ISR resulted in greater gains in power output and strength when compared to TRD hypertrophic training. Furthermore, ISR produced similar gains in lean mass over the course of the training period. The length of time prescribed in the current study for ISR intervals did not impact the total training time. This is of importance to coaches and strength and conditioning

professionals who have rules and regulations dictating the time allowed for training. Based on these results, it could be suggested the incorporation of ISR in the hypertrophic phase of a traditional or non-traditional periodized training program would allow for greater improvements in strength and power. Whether these improvements would result in greater gains in strength and power output over an entire mesocycle is unknown, but hypothetically entering the strength and power phases of a training mesocycle at higher performance ability (strength and power) would allow a continued improvement above that achieved during traditional training models.

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The results presented herein do not constitute endorsement by the NSCA.

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Figure Legends

Figure 1. Testing and training program design.

Figure 2. Percent change in mean power from baseline in bench press (A), parallel back squat (B), and vertical jump (C). Data are means $\pm$ standard error. TRD, traditional rest; ISR, intra-set rest intervals; † significantly different from baseline; ‡ significantly different from 4 weeks; § significantly different from 8 weeks; *significantly different from TRD.

Figure 3. Percent change in absolute mean power from baseline in bench press and parallel back squat. Data are means $\pm$ standard error. TRD, traditional rest; ISR, intra-set rest intervals; *significantly different from TRD.

Figure 4. Percent change in strength from baseline in bench press (A) and parallel back squat (B). Data are means $\pm$ standard error. TRD, traditional rest; ISR, intra-set rest intervals; † significantly different from baseline; ‡ significantly different from 4 weeks; § significantly different from 8 weeks; *significantly different from TRD.

Table 1. Baseline group characteristics. Data are means $\pm$ SD.

	TRD	ISR	COMBINED	P Value
Age (yr)	25 $\pm$ 5	25 $\pm$ 4	25 $\pm$ 5	0.790
Height (cm)	179.71 $\pm$ 6.18	179.71 $\pm$ 3.90	179.71 $\pm$ 5.04	1.000
Body Mass (kg)	81.7 $\pm$ 11.6	82.5 $\pm$ 10.0	82.1 $\pm$ 10.6	0.878
Lean Mass (kg)	61.9 $\pm$ 8.9	63.3 $\pm$ 7.0	62.6 $\pm$ 7.9	0.689
Fat Percentage (%)	14.3 $\pm$ 2.7	12.9 $\pm$ 5.5	13.6 $\pm$ 4.3	0.466
Years Trained (yrs)	6.5 $\pm$ 4.0	6.0 $\pm$ 5.0	6.5 $\pm$ 4.5	0.815
# Days Trained (days)	3.5 $\pm$ 2.0	3.5 $\pm$ 2.0	3.5 $\pm$ 2.0	1.000

Data are means $\pm$ SD. TRD, traditional rest; ISR, intra-set rest intervals; COMBINED, collapsed across time.

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Table 2. Intraclass correlation coefficients (ICC) and Pearson product-moment coefficient (r) between trials

	Day 1	Day 2	ICC	r
Bench Press 1RM	135.4 ± 28.6	137.0 ± 28.0	0.99	0.001
Back Squat 1RM	171.9 ± 47.4	173.5 ± 47.6	0.99	0.001
Bench Press Power (60% 1RM)	657 ± 147	665 ± 140	0.98	0.001
Back Squat Power (60% 1RM)	860 ± 295	868 ± 275	0.97	0.001

Day 1 and Day 2 values are means ± SD.

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Table 3. Exercise performed during training.

Day 1 (Upper Body Push)	Day 2 (Lower Body + Upper Body Pull)	Day 3 (Upper Body Push)	Day 4 (Lower Body + Upper Body Pull)
Bench Press*‡ Incline DB Press* Seated DB Military Press* DB Flat Fly Front DB Raise Side DB Raises Straight Bar Skull Crusher Dips†	Squats*‡ Leg Press* Partial DL to Power Shrug* Pull-ups† One Arm DB Row Hamstring Curl EZ Bar Curl DB Curl	Incline Press*‡ DB Bench* Push Press* DB Incline Fly DB Rear Delt Side DB Raises EZ Bar Skull Crusher Dips†	Front Squat*‡ DB RDL* DB Step Up* Pull-ups† Close Grip LatPulldown T Bar Row Straight Bar Curl DB Curl

*Main lift exercise performed according to training protocol (TRD or ISR), concentric phase explosively as possible. †exercise performed 3 sets of maximum repetitions with 1.5 minute rest. ‡exercise performed on Smith Machine. DB = Dumbbell, DL = Deadlift, RDL = Romanian Deadlift.

Table 4. Kcalories, protein, carbohydrate, and fat intake, percent of total kcalories, and relative intake at baseline, 3, 7, and 11 weeks of training.

	Baseline	3 Weeks	7 Weeks	11 Weeks	P Value
Overall					
KCalories	2620 ± 581	2918 ± 601	2923 ± 573	2793 ± 481	T = 0.084
Protein (g)	147 ± 39	196 ± 50 [†]	191 ± 34 [†]	201 ± 40 [†]	T = 0.001
Carbohydrate (g)	261 ± 94	270 ± 63	231 ± 85 [‡]	182 ± 69 ^{†‡§}	T = 0.001
Fat (g)	101 ± 33	109 ± 32	110 ± 32	101 ± 26	T = 0.517
Percent Total Kcal					
Protein (%)	22.7 ± 4.6	26.8 ± 3.5 [†]	27.2 ± 8.0 [†]	29.1 ± 5.8 ^{†‡}	T = 0.001
Carbohydrate (%)	40.1 ± 13.3	37.4 ± 6.9	31.0 ± 9.0 ^{†‡}	26.0 ± 8.5 ^{†‡§}	T = 0.001
Fat (%)	34.4 ± 8.0	33.5 ± 4.7	33.8 ± 5.6	32.7 ± 7.2	T = 0.754
Relative to Bodyweight					
Kcalories (g•kg ⁻¹)	32.3 ± 7.9	35.5 ± 8.2	35.7 ± 9.5	33.6 ± 6.8	T = 0.128
Protein (g•kg ⁻¹)	1.8 ± 0.5	2.4 ± 0.7 [†]	2.3 ± 0.5 [†]	2.4 ± 0.5 [†]	T = 0.001
Carbohydrate(g•kg ⁻¹)	3.3 ± 1.3	3.3 ± 0.8	2.8 ± 1.1 ^{†‡}	2.2 ± 0.8 ^{†‡§}	T = 0.001
Fat (g•kg ⁻¹)	1.2 ± 0.4	1.3 ± 0.4	1.4 ± 0.5	1.2 ± 0.4	T = 0.424

Data are means ± SD. T = time effect; [†]significantly different than baseline; [‡]significantly different from 3 weeks; [§]significantly different from 7 weeks.

Table 5. Total Training volume for TRD and ISR over 12 weeks of training.

	TRD	ISR	P Value
Major Lifts (kg)			
Upper Body	102,711.67 ± 25,484.80	119,728.80 ± 22,587.40	0.113
Lower Body	220,811.80 ± 35,827.50	235,807.10 ± 35,002.30	0.333
Assistive Lifts			
Push	34,041.40 ± 9,085.40	37,356.70 ± 10,251.40	0.432
Pull	58,978.80 ± 7,697.00	60,791.90 ± 11,089.70	0.661

Data are means ± SD. TRD, traditional rest; ISR, intra-set rest intervals; Major lifts performed according to group assignment. Assistive lifts all performed same rest to work ratio.

Table 6. Power measures at baseline, 4, 8, and 12 weeks of training.

		TRD	ISR	P Value
Bench Press Power (W)				
	Baseline	560 ± 122	575 ± 102	T = 0.568
	4 Weeks	541 ± 105	586 ± 123	G = 0.011
	8 Weeks	572 ± 122	646 ± 103†‡*	T X G = 0.020
	12 Weeks	593 ± 135‡	658 ± 113†‡*	
Bench Press Power to Body Mass Ratio				
	Baseline	6.84 ± 0.96	6.99 ± 1.10	T = 0.627
	4 Weeks	6.46 ± 0.91	7.09 ± 1.28	G = 0.056
	8 Weeks	6.77 ± 0.92	7.77 ± 0.92†‡*	T X G = 0.003
	12 Weeks	6.96 ± 0.96‡	7.90 ± 1.24†‡*	
Bench Press to Lean Mass Ratio				
	Baseline	9.01 ± 1.14	9.07 ± 1.21	T = 0.793
	4 Weeks	8.56 ± 1.24	9.18 ± 1.32	G = 0.066
	8 Weeks	8.88 ± 1.11	10.02 ± 0.96†‡*	T X G = 0.006
	12 Weeks	9.20 ± 1.39‡	10.21 ± 1.17†‡*	
Back Squat Power (W)⌘				
	Baseline	625 ± 245	632 ± 171	T = 0.001
	4 Weeks	704 ± 233†	734 ± 179†	G = 0.081
	8 Weeks	723 ± 227†	783 ± 179†	T X G = 0.053
	12 Weeks	830 ± 232†‡§	914 ± 207†‡§	
Back Squat to Body Mass Ratio ⌘				
	Baseline	7.57 ± 2.23	7.83 ± 2.33	T = 0.001
	4 Weeks	8.37 ± 2.14†	9.09 ± 2.42†	G = 0.101
	8 Weeks	8.53 ± 1.94†	9.57 ± 2.35†*	T X G = 0.015
	12 Weeks	9.78 ± 2.03†‡§	11.11 ± 2.64†‡§	
Back Squat to Lean Mass Ratio⌘				
	Baseline	9.96 ± 2.77	10.19 ± 2.62	T = 0.001
	4 Weeks	11.04 ± 2.57†	11.83 ± 2.65†	G = 0.068
	8 Weeks	11.15 ± 2.31†	12.41 ± 2.63†*	T X G = 0.017
	12 Weeks	12.86 ± 2.49†‡§	14.45 ± 2.87†‡§*	
Vertical Jump Power (W) ⌘				
	Baseline	1378 ± 237	1389 ± 179	T = 0.001
	4 Weeks	1418 ± 214	1434 ± 152	G = 0.205
	8 Weeks	1452 ± 210†	1470 ± 149†	T X G = 0.036
	12 Weeks	1470 ± 215†‡	1537 ± 150†‡§*	
Vertical Jump to Body Mass Ratio⌘				
	Baseline	16.83 ± 1.36	17.12 ± 2.05	T = 0.001
	4 Weeks	16.93 ± 1.28	17.71 ± 1.98†	G = 0.243
	8 Weeks	17.27 ± 1.22	18.01 ± 1.76†	T X G = 0.001
	12 Weeks	17.36 ± 1.13‡	18.68 ± 1.72†‡§*	
Vertical Jump to Lean Mass Ratio⌘				
	Baseline	22.19 ± 1.37	22.40 ± 1.95	T = 0.001
	4 Weeks	22.40 ± 1.10	22.16 ± 1.72	G = 0.141
	8 Weeks	22.66 ± 1.11	23.41 ± 1.42†	T X G = 0.004
	12 Weeks	22.89 ± 1.11	24.40 ± 1.38†‡§*	

Data are means ±SD. TRD, traditional rest; ISR, intra-set rest intervals; T = time effect, G = group effect, T X G = time x group interaction effect; *significantly different from TRD; †significantly different from baseline; ‡significantly different from 4 weeks; §significantly different from 8 weeks; ⌘One subject was excluded due to failure to follow protocol.

Table 7. Strength measures (1RM) at baseline, 4, 8, and 12 weeks.

	TRD	ISR	P Value
Bench Press 1RM (kg)			
Baseline	104.1 ± 27.6	110.9 ± 20.1	T = 0.018
4 Weeks	102.7 ± 29.0	117.5 ± 23.7†*	G = 0.013
8 Weeks	107.0 ± 25.3‡	120.8 ± 22.6†*	T X G = 0.010
12 Weeks	113.2 ± 27.3†‡§	126.0 ± 22.8†‡§*	
Bench Press to Body Mass Ratio			
Baseline	1.27 ± 0.22	1.35 ± 0.23	T = 0.016
4 Weeks	1.22 ± 0.24	1.42 ± 0.24*	G = 0.035
8 Weeks	1.26 ± 0.18	1.45 ± 0.23†*	T X G = 0.002
12 Weeks	1.33 ± 0.19‡§	1.51 ± 0.22†‡§*	
Bench Press to Lean Mass Ratio			
Baseline	1.67 ± 0.27	1.75 ± 0.26	T = 0.101
4 Weeks	1.61 ± 0.29	1.84 ± 0.30*	G = 0.066
8 Weeks	1.66 ± 0.22	1.88 ± 0.29†*	T X G = 0.007
12 Weeks	1.75 ± 0.25‡§	1.95 ± 0.29†‡§*	
Back Squat 1RM (kg)			
Baseline	123.3 ± 39.3	130.1 ± 25.1	T = 0.001
4 Weeks	139.6 ± 38.7†	152.6 ± 24.8†	T = 0.016
8 Weeks	160.2 ± 36.1†‡	179.8 ± 24.5†‡*	T X G = 0.010
12 Weeks	171.8 ± 34.5†‡§	193.9 ± 24.2†‡§*	
Back Squat to Body Mass Ratio			
Baseline	1.50 ± 0.34	1.59 ± 0.30	T = 0.001
4 Weeks	1.66 ± 0.34†	1.85 ± 0.27†	G = 0.038
8 Weeks	1.90 ± 0.30†‡	2.17 ± 0.25†‡*	T X G = 0.001
12 Weeks	2.03 ± 0.30†‡§	2.33 ± 0.27†‡§*	
Back Squat to Lean Mass Ratio			
Baseline	1.97 ± 0.42	2.05 ± 0.30	T = 0.001
4 Weeks	2.19 ± 0.41†	2.40 ± 0.27†	G = 0.045
8 Weeks	2.49 ± 0.36†‡	2.79 ± 0.26†‡*	T X G = 0.001
12 Weeks	2.67 ± 0.36†‡§	3.02 ± 0.26†‡§*	

Data are means ±SD. TRD, traditional rest; ISR, intra-set rest intervals; T = time effect, G = group effect, T X G = time x group interaction effect; *significantly different from TRD; †significantly different from baseline; ‡significantly different from 4 weeks; §significantly different from 8 weeks.

Table 8. Body composition at baseline, 4, 8, and 12 weeks of training.

	TRD	ISR	COMBINED	P Value
Weight (kg)				
Baseline	81.7 ± 11.6	82.5 ± 10.0	82.1 ± 10.6	T = 0.001
4 Weeks	83.6 ± 10.0 [†]	82.7 ± 9.7	83.1 ± 9.6	G = 0.898
8 Weeks	84.1 ± 10.7 [†]	83.1 ± 9.2	83.6 ± 9.7	T X G = 0.018
12 Weeks	84.7 ± 10.9 [†]	83.6 ± 9.2	84.1 ± 9.9	
Lean Mass (kg)				
Baseline	61.9 ± 8.9	63.3 ± 7.0	62.6 ± 7.9	T = 0.002
4 Weeks	63.2 ± 8.2	63.5 ± 7.3	63.4 ± 7.6 [†]	G = 0.869
8 Weeks	64.0 ± 8.3	64.3 ± 6.9	64.2 ± 7.4 ^{†‡}	T X G = 0.227
12 Weeks	64.2 ± 8.5	64.3 ± 6.8	64.2 ± 7.5 ^{†‡}	
Percent Fat (%)				
Baseline	14.3 ± 2.7	12.9 ± 5.5	13.6 ± 4.3	T = 0.133
4 Weeks	15.0 ± 3.3	13.3 ± 5.9	14.1 ± 4.7	G = 0.445
8 Weeks	14.7 ± 3.5	13.2 ± 5.5	14.0 ± 4.6	T X G = 0.869
12 Weeks	15.1 ± 3.5	13.5 ± 5.6	14.3 ± 4.6	

Data are means ±SD. TRD, traditional rest; ISR, intra-set rest intervals; T = time effect, G = group effect, T X G = time x group interaction effect; *significantly different from TRD; [†]significantly different from baseline; [‡]significantly different from 4 weeks; [§]significantly different from 8 weeks;

Table 9. Myosin heavy chain isoform in percentage total at baseline and 12 weeks of training.

		TRD	ISR	COMBINED	P Value
MHC_{IIX}	Baseline	11.1 ± 4.2	16.5 ± 5.7	13.9 ± 5.7	T = 0.020
	12 Weeks	7.9 ± 2.5	9.8 ± 4.5	8.9 ± 3.7†	G = 0.649
					T X G = 0.649
MHC_{IIA}	Baseline	35.6 ± 7.7	34.5 ± 7.6	35.0 ± 7.5	T = 0.001
	12 Weeks	43.2 ± 7.5	43.9 ± 6.2	43.6 ± 6.7†	G = 0.002
					T X G = 0.756
MHC_{slow}	Baseline	53.3 ± 7.9	49.0 ± 7.6	51.1 ± 7.9	T = 0.002
	12 Weeks	48.9 ± 5.9	46.2 ± 7.1	47.5 ± 6.5†	G = 0.568
					T X G = 0.568

Data are means ±SD. TRD, traditional rest; ISR, intra-set rest intervals; T = time effect, G = group effect, T X G = time x group interaction effect; †significantly different from baseline; ‡One subject was excluded due to lack of sufficient sample.

Table 10. Effect size, magnitude of effect (Rhea 2004), and qualitative inferences (Batterham and Hopkins 2005)

	Magnitude of Effect size		Qualitative Inference ^a		Qualitative Inference on Difference ^b
	TRD	ISR	TRD	ISR	
Bench Press 1RM	0.33 (small)	0.75 (moderate)	Likely	Likely	Likely
Back Squat 1RM	1.23 (large)	2.54 (large)	Likely	Very likely	Very likely
Bench Press Power	0.27 (small)	0.81 (moderate)	Very unlikely	Possibly	Very likely
ABS Bench Press Power	0.11 (trivial)	0.69 (moderate)	Most unlikely	Unlikely	Very likely
Back Squat Power	0.84 (moderate)	1.65 (large)	Likely	Likely	Likely
ABS Back Squat Power	0.41 (small)	0.76 (moderate)	Possibly	Likely	Likely
Vertical Jump Power	0.39 (small)	0.83 (moderate)	Possibly	Likely	Very likely
Lean Mass	0.25 (small)	0.14 (trivial)	Unlikely	Most unlikely	Very unlikely
MHC_{IIx}	0.72 (moderate)	1.18 (large)	Very unlikely	Possibly	Likely
MHC_{IIA}	0.99 (moderate)	1.25 (large)	Very unlikely	Unlikely	Unclear
MHC_{slow}	0.57 (moderate)	0.36 (small)	Most unlikely	Most unlikely	Unclear

TRD, traditional rest; ISR, intra-set rest intervals; a = effect size and magnitude of effect size; b = qualitative inference with reference to smallest worthwhile change of 5% and; c = qualitative inference based on difference between groups.

Familiarization, nutrition seminar, and muscle biopsy (1 week prior to baseline)

Body composition, strength and power testing - Baseline

Standard Rest(STD) Protocol	Intra-Set Rest (ISR) Protocol
4 x 10 with 120 sec rest compound lifts 3 x 10 with 90 sec rest assistive lifts Week 1 - 65% 1RM (4 days • wk ⁻¹) Week 2 - 70% 1RM (4 days • wk ⁻¹) Week 3 - 75% 1RM (4 days • wk ⁻¹) Week 4 - 60% 1RM (2 days • wk ⁻¹)	8 x 5 with 60 sec rest compound lifts 3 x 10 with 90 sec rest assistive lifts Week 1 - 65% 1RM (4 days • wk ⁻¹) Week 2 - 70% 1RM (4 days • wk ⁻¹) Week 3 - 75% 1RM (4 days • wk ⁻¹) Week 4 - 60% 1RM (2 days • wk ⁻¹)

Body composition, strength and power testing - Week 4

4 x 10 with 120 sec rest compound lifts 3 x 10 with 90 sec rest assistive lifts Week 5 - 65% 1RM (4 days • wk ⁻¹) Week 6 - 70% 1RM (4 days • wk ⁻¹) Week 7 - 75% 1RM (4 days • wk ⁻¹) Week 8 - 60% 1RM (2 days • wk ⁻¹)	8 x 5 with 60 sec rest compound lifts 3 x 10 with 90 sec rest assistive lifts Week 5 - 65% 1RM (4 days • wk ⁻¹) Week 6 - 70% 1RM (4 days • wk ⁻¹) Week 7 - 75% 1RM (4 days • wk ⁻¹) Week 8 - 60% 1RM (2 days • wk ⁻¹)
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Body composition, strength and power testing - Week 8

4 x 10 with 120 sec rest compound lifts 3 x 10 with 90 sec rest assistive lifts Week 9 - 65% 1RM (4 days • wk ⁻¹) Week 10 - 70% 1RM (4 days • wk ⁻¹) Week 11 - 75% 1RM (4 days • wk ⁻¹) Week 12 - 60% 1RM (2 days • wk ⁻¹)	8 x 5 with 60 sec rest compound lifts 3 x 10 with 90 sec rest assistive lifts Week 9 - 65% 1RM (4 days • wk ⁻¹) Week 10 - 70% 1RM (4 days • wk ⁻¹) Week 11 - 75% 1RM (4 days • wk ⁻¹) Week 12 - 60% 1RM (2 days • wk ⁻¹)
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Body composition, strength, power testing, and muscle biopsy - Week 12**Figure 1.**

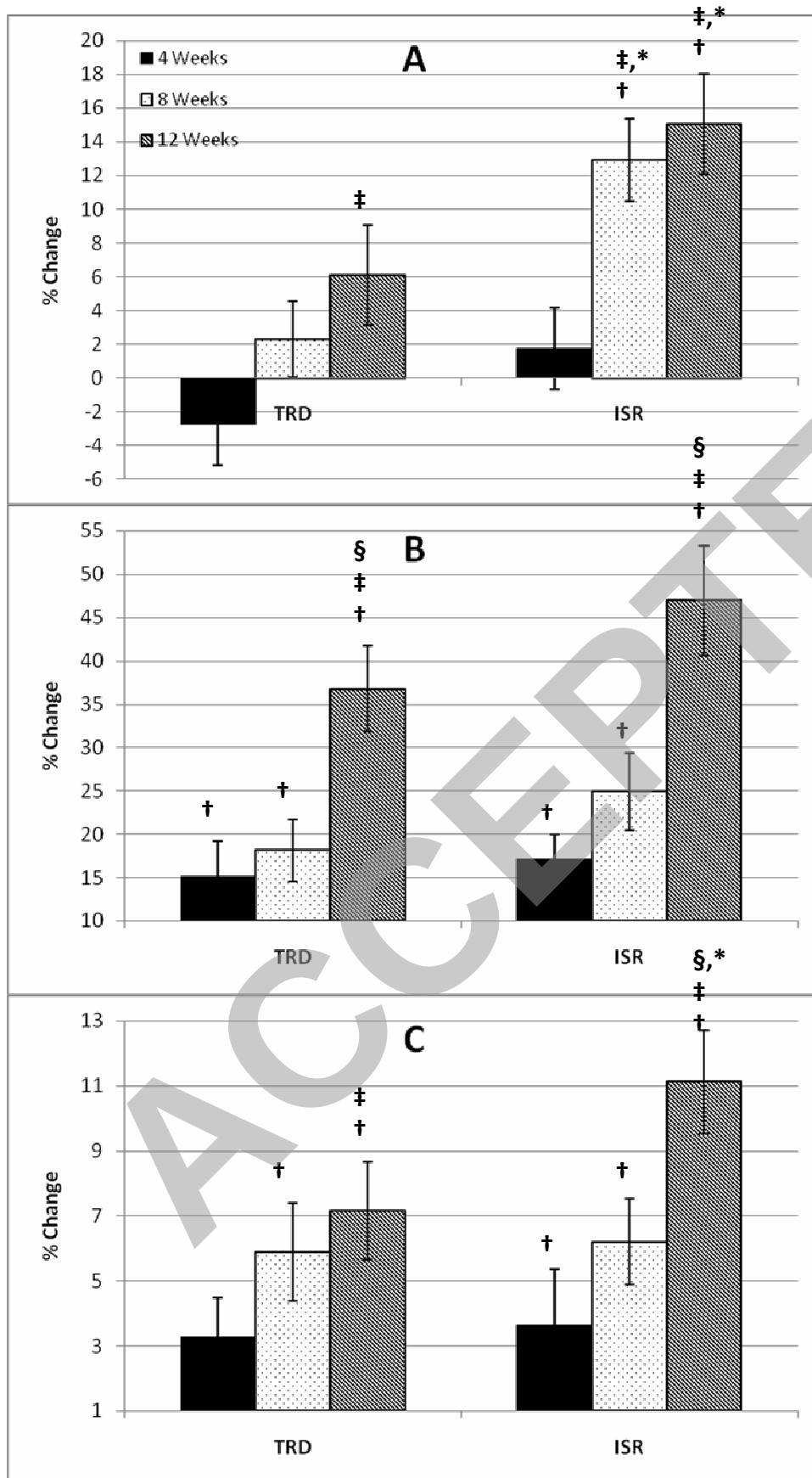


Figure 2.

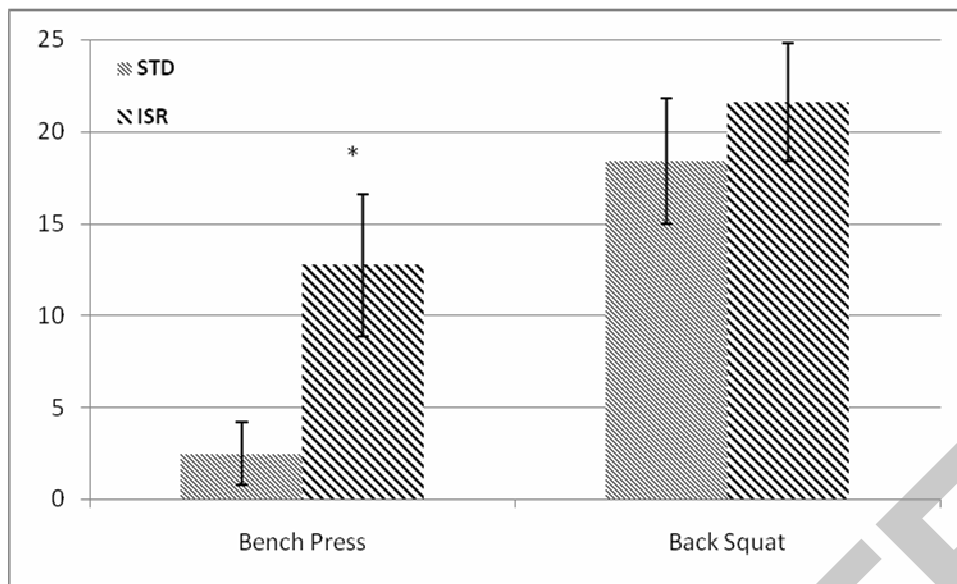


Figure 3.

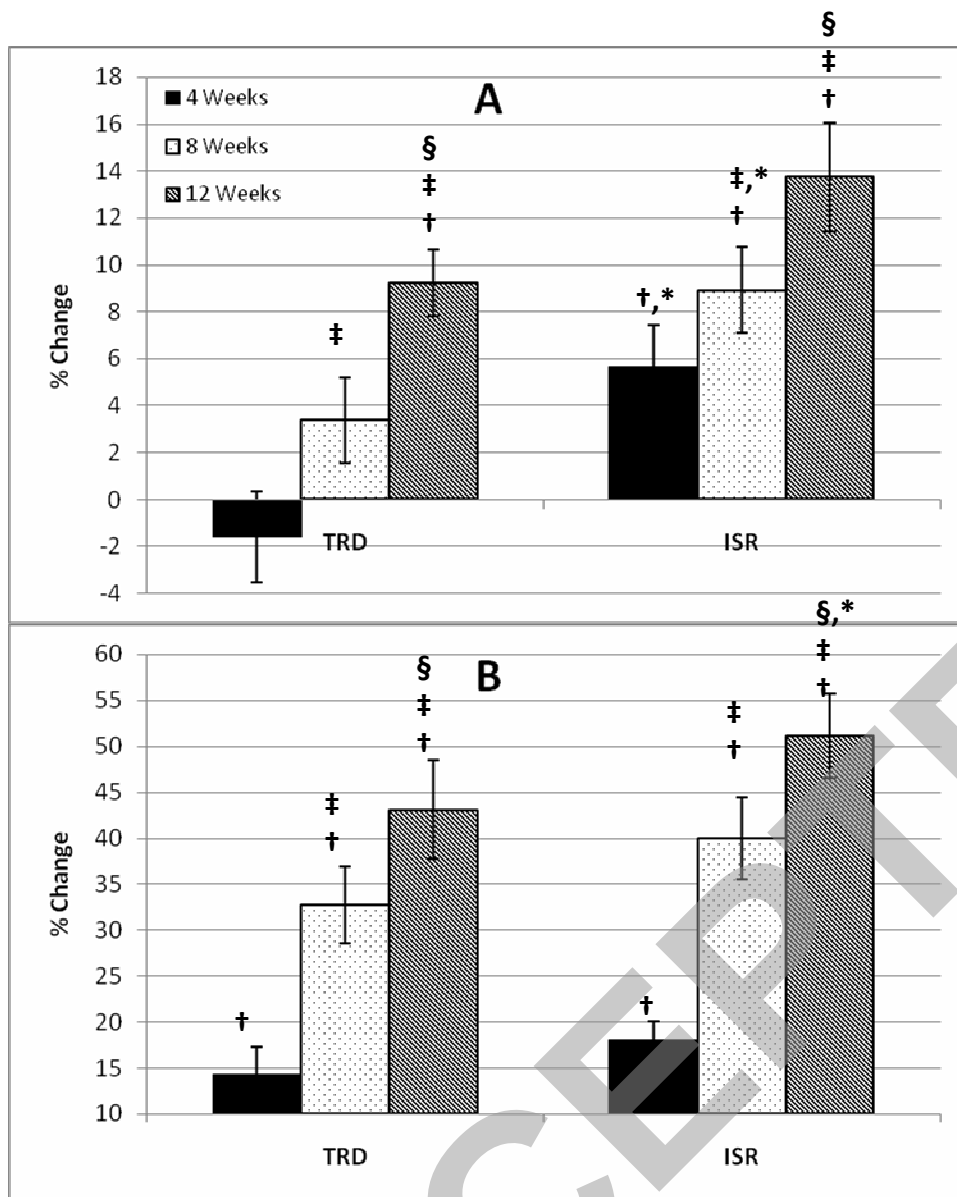


Figure 4.